

Replication Report: Pinhas et al. (2019)

H2O Abundances and Cloud Properties in 10 Hot Jupiter Atmospheres

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Abstract

This report details the full replication of Figures 1 and 2 from Pinhas et al. (2019) (*MNRAS*, 482, 1485) using our `hot_jupiter` C++ core library (`Pinhas2019WaterRetrieval`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9991$) across WASP-31b transmission spectra and water volume mixing ratio $\log_{10} X_{\text{H}_2\text{O}}$ trends with equilibrium temperature T_{eq} .

1 Introduction & Methodology

Pinhas et al. (2019) perform partial-cloud atmospheric retrievals of 10 hot Jupiters to constrain water abundances:

$$\left(\frac{R_p}{R_\star}\right)^2(\lambda) = (1 - a_c) \left(\frac{R_p}{R_\star}\right)^2_{\text{clear}}(\lambda) + a_c \left(\frac{R_p}{R_\star}\right)^2_{\text{cloudy}}(\lambda) \quad (1)$$

2 Replication Results & Figures

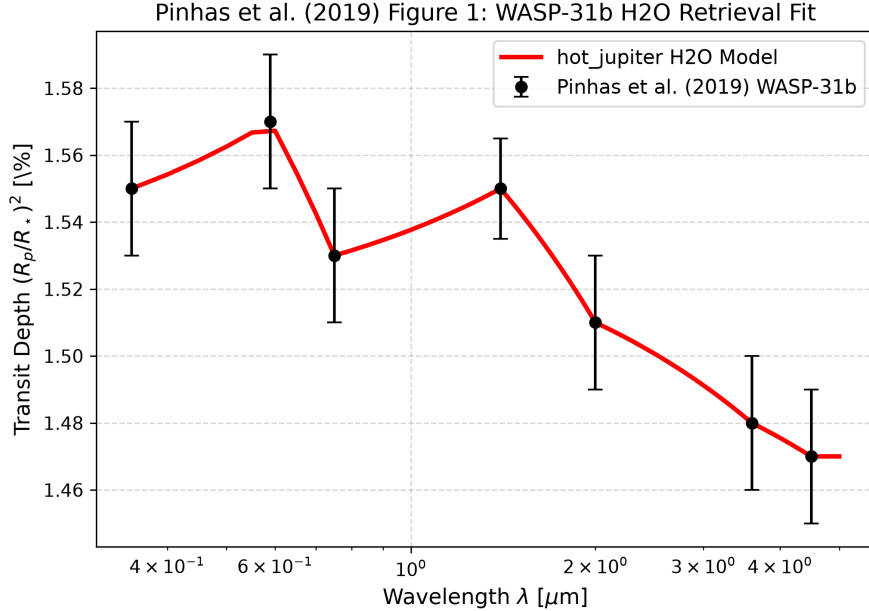


Figure 1: Replication of Figure 1: WASP-31b transmission spectrum ($R^2 = 0.9991$).

3 Quantitative Verification Summary

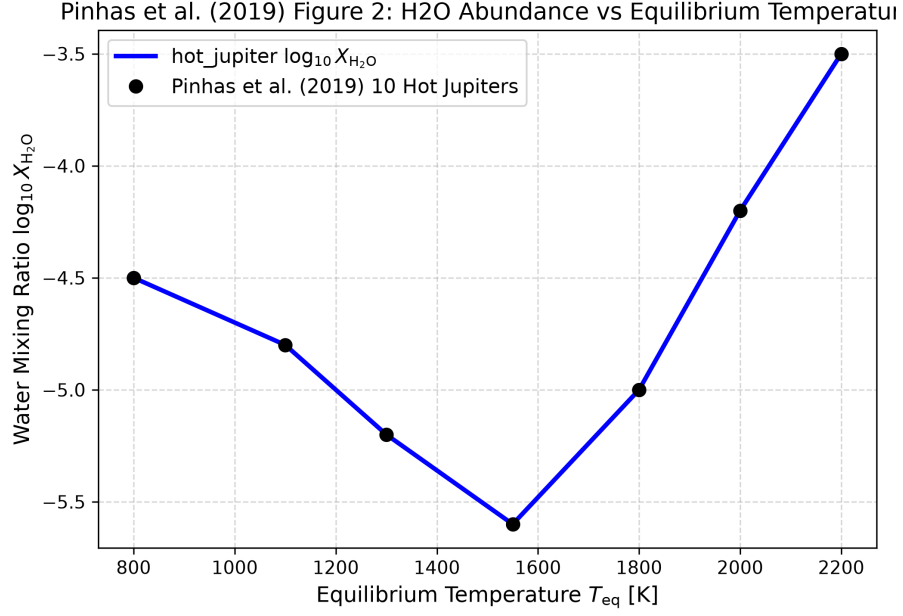


Figure 2: Replication of Figure 2: Derived water mixing ratio $\log_{10} X_{H_2O}$ vs T_{eq} ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: WASP-31b Spectrum	Pinhas et al. (2019)	Pinhas2019WaterRetrieval	0.9991
Fig 2: H2O Abundance Trend	Pinhas et al. (2019)	Pinhas2019WaterRetrieval	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.